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Artificial intelligence to optimize water consumption in agriculture: A predictive algorithm-based irrigation management system

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ABSTRACT

Water scarcity is a pressing global issue that needs to be faced. The United Nations highlights that only about 31 percent of the population is not characterized by water stress, meaning that the world's freshwater resources are unevenly distributed and unsustainably managed. This phenomenon is both natural and man-made, as the human footprint is linked to many disciplines, with the largest impact coming from the agricultural sector. Scientific literature highlights how 4.0 technologies are the key drivers to reduce this sector's impact on valuable resources, such as water and soil.

Based on these premises, the proposed work aims at improving the irrigation management in agriculture by implementing a three-layer architecture system to optimize water consumption and prevent soil percolation, by avoiding situations where the soil moisture exceeds its capacity point. To achieve this, experimental activities allow the evaluation of the soil capacity point and thus the definition of a confidence interval to guide watering decisions. The latter interval, soil and environmental data, and three-day weather forecasts are aggregate to create a consistent dataset for training and testing three different machine learning algorithms based on a classification problem to predict the state of the irrigation network.

As a result, the implemented multi-layer perceptron neural network, support vector machine, and k-neighbors classifier achieved an accuracy of nearly 99%. Despite this, the neural network produced superior decision region boundaries, resulting in fewer false predictions. A Monte Carlo simulation was then applied to evaluate the water and energy savings, which were up to 27 % and 57 %, respectively.

In summary, the predictive algorithm-based irrigation management system is a cost-effective solution for optimizing water management in agriculture that it is truly scalable to any crop by assessing the appropriate soil capacity level.

1. Introduction

Water scarcity is a critical challenge for sustainable development that is truly influenced by the human footprint and climate change. According to the United Nations, a state of water stress is defined as a 25 % of water withdrawal from the renewable freshwater resources. Sustainable Development Goal indicator 6.4.2 states that in 2018, 18.4 % of the global renewable freshwater was withdrawn by all economic activities (United Nations, 2024). One of the largest contributions is strongly related to agricultural activities, as the irrigation management accounts for nearly 70 % of the global water withdrawals (Food and Agriculture Organization of the United Nations, 2023). In this framework, future research aims at addressing the water scarcity issue from different action points. Therefore, 4.0 technologies are emerging as a potential solution to improve the sustainability of the agricultural sector with multiple sources, such as blockchain, internet of things technologies (IoT), deep learning and machine learning algorithms, and other computer applications (Preite et al., 2023b). Predictive algorithms are straightforward solutions widely used in a 4.0 scenario across multiple disciplines because they provide highly scalable automation (Mazzei & Ramjattan, 2022). The scientific literature shows a few of machine learning approaches in agriculture, which can be categorized into three groups: pre-harvesting, harvesting, and post harvesting (Liakos et al., 2018, Meshram et al., 2021). Applications identified for irrigation purposes, such as water scarcity recognition, water demand predictions, and irrigation scheduling were in the first group. The irrigation is traditionally scheduled by considering a fixed period and overlooking the variability in environmental and plant properties. Specifically, water scarcity

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recognition deals with the evaluation of the drought stress, stem water potential and the plant water content by processing thermal infrared images, meteorological and soil data. In this case, convolutional neural network, random forest, and gradient-boosted decision trees were the main employed models (Zhou et al., 2021). Multispectral and thermal images in combination with meteorological and soil data were also used to estimate the soil moisture content, reference evapotranspiration and sap flow properties by implementing support vector machine, artificial neural network, and gradient-boosting decision trees algorithms (Alibabaei et al., 2021). The authors of (Corell et al., 2016) provide an irrigation framework based on sensors data for comparing three regressor models to determine the suitable irrigation volume for an olive cultivation. Corn, kiwi, and potato crops were monitored to implement a fuzzy decision support system for evaluating the proper irrigation amount by considering the growing degree days, the amount of water delivered to the plants, and the evapotranspiration rate (Giusti & Marsili-Libelli, 2015). A partial least-square regression and an adaptive neuro fuzzy inference system were applied to provide watering recommendations for lemon trees by processing soil moisture, evapotranspiration, and humidity (Navarro-Hellín et al., 2016). Soil moisture trends in both vertical depth and time were assessed by (Chandrappa et al., 2022), which develops both machine learning (support vector regression and linear regression) and deep learning algorithms (long short-term memory). In this background, a significant correlation between wind speed and soil moisture in multi depth was highlighted. A water saving of 20 % was achieved by (Gu et al., 2021) developing an artificial neural network capable to determine the irrigation duration working on data provided by soil sensors and weather station. Artificial intelligence application was also explored by (Kavya et al., 2023) for predicting short term water demand. Specifically, the predictive performance of the machine learning and deep learning was evaluated with both univariate and multivariate time series. The univariate series has been implemented considering only the flow meter data, meanwhile meteorological data were also considered for the multivariate scenario. Time series forecasting to estimate soil moisture for irrigation purposes has been also explored by (Custódio & Prati, 2024), which compares the performance of modern and traditional modeling methods by testing three different scenarios. The first aimed at gaining insight into the impact of the training window size on the prediction performance of the multivariate algorithms. In the second scenario, the same test was repeated but the frequency of fitting the algorithms was modified, and finally, in the third scenario, the most performing multivariate algorithm was compared with univariate algorithms by evaluating ten datasets from different regions of the world. As a result, the authors stated that lower prediction capabilities resulted with algorithms trained with short-period samples. Furthermore, multivariate random forest, and extreme gradient boosting showed higher performance than statistical methods, such as vector autoregression. Finally, the univariate methods outperformed the multivariate methods, suggesting that exogenous variables have limited influence in predicting soil moisture time series data. In (Mokhtar et al., 2023), different machine learning algorithms were developed to evaluate irrigation water requirements starting from the maximum and minimum temperature, relative humidity, wind speed, precipitation, root depth, basal crop coefficient, evapotranspiration, and the fraction of the wetted and exposed soil. High performance was reached with the convolutional neural network and the long short-term memory, meanwhile the support vector machine resulted in the lowest accuracy. The authors of (Srivastava et al., 2024) developed a probabilistic framework to determine irrigation strategies based on three different parameters: soil moisture, leaf area index, and evapotranspiration. The described indicators reflect the soil water deficit, crop water stress, and the water demand, respectively. In this framework, a random forest regressor was used to determine suitable predictors for each parameter, while side the predictions were entrusted to a recurrent artificial neural network (long short-term memory). Finally, the resulting weights were adjusted by comparing the predicted and actual values.

Evapotranspiration is an important variable for implementing accurate irrigation decisions, which reflects both the metabolic activities of the plant and the meteorological data. The authors of (Aly et al., 2024) exploited extra tree regressor, support vector regressor, k-nearest neighbor and AdaBoost regression for feeding data into a super learner ensemble to estimate the reference evapotranspiration, achieving high accuracy with limited meteorological data. The latter issue was also highlighted by (Yong et al., 2024) as the main limitation for predicting evapotranspiration rate, pointing out a hybrid neuro-fuzzy inference system for counteract this problem. In this perspective, the author of (Adnan et al., 2021) investigated the applicability of hybrid support vector regression models, which combine machine learning techniques with optimization *meta*-heuristic algorithms (i.e., Particle Swarm Optimization, Whale Optimization, Differential Evolution, and Covariance Matrix Adaptation Evolution Strategy). Following the same logical path, (Považanová et al., 2023) improved prediction accuracy by combining machine learning and feature engineering for estimating reference evapotranspiration, providing valuable insight into the effectiveness and the generalization of the proposed models. The authors of (Youssef et al., 2024) showed how several machine learning algorithms, such as k-nearest neighbors, support vector machine, multilinear regression, and decision tree achieved an accuracy close to 99 % in estimating reference evapotranspiration.

The scientific literature also outlines the main challenges of applying machine learning to irrigation. First, these applications require high quality data sources for ensuring an effective learning rate and accurate predictions. Thereby implementing a dataset for an entire irrigation season is challenging because the growth period is truly long, and many parameters influence the crop quality and water demand. In this context, scalability is the other pressing issue as many predictive models fail when crop, soil, environmental and irrigation conditions differ (Sharma et al., 2021).

The main aim of this study is to address the water shortage issue by proposing a machine learning application to schedule the irrigation by processing soil, environmental and weather forecast data. The experimental activities were carried out in a living lab developed within the framework of the National Research Center for Technology in Agriculture. The living lab focused on a tomato crop (Solanum Lycoperium L. Cv. HEINZ 1301) irrigated according to the watering recommendation provided by the IRRIFRAME platform, which amounts to 324.5 mm for a cultivated surface of 0.0132 ha (Irriframe, 2023). In this framework, a LoRaWAN-based IoT network has been installed to real-time monitoring soil and environmental conditions and controlling the water delivery. The collected data were aggregate with the three-day weather forecast for implementing a dataset to train and validate a predictive algorithms-based irrigation management system able to avoid overwatering by maintaining an adequate soil moisture level, which ranges around the soil capacity point. In this framework, the soil capacity point is experimentally evaluated by computing the time derivative of the soil moisture to identify the area where the rate of decline is slower after an irrigation or a precipitation event. From this perspective, an irrigation confidence interval has been defined, which allows water and energy consumptions to be minimized to increase the sustainability of the irrigation at the farm level. Specifically, the proposed framework responds to a multi-classification problem, where the outputs are used to trigger the control valves or to send warning messages requiring farmer actions. Deeping into the technical features, three different supervised models have been selected by considering the most investigated applications in the scientific literature previously described. Specifically, two of the most widely used machine learning algorithms for classification purposes, k-nearest neighbors, support vector machine, were compared with a multi-layer perceptron artificial neural network to define the more suitable solution. Therefore, the first has been selected for its flexibility in handling different types of data, but it was highly sensitive to outliers. Support vector machine is more complicated to train for multi-classification problems, but it provides robustness in dealing with

outliers and avoiding overfitting. In addition, it encompasses kernel functions to increase its effectiveness in high-dimensional spaces. Finally, multi-layer perceptron neural network is one of the most widespread feedforward deep learning algorithms due to its reliability and its capability to maximize the accuracy by using a back-propagation algorithm during the training phase (Hemalatha K. et al., 2017). Prediction performance is evaluated by defining the resulting decision boundary regions, the confusion matrix and calculating various indicators, such as accuracy, precision, recall, F1-score, the corresponding macro-average precision, and the weighted-average precision.

Summarizing the results, all the models associate the classification label to unknown data with high degree of accuracy, close to 99 %. Although the high accuracy achieved by the three algorithms, significant differences are highlighted when considering the resulting decision boundaries and the resulting confusion matrix. Indeed, the artificial neural network shows more effective decision regions that allow to minimize false predictions. In addition, this framework can be truly scaled to other crops and soil conditions since the new operating conditions can be simulated by evaluating the new soil capacity point. Water and energy savings are determined by performing a Monte Carlo simulation of 1000 tomato seasons and comparing the results with the amount of water recommended by the IRRIFRAME platform. The first one ranges from 14.5 % to 27.6 %, while the second ranges from 49.2 % to 57 %.

The methodology adopted for the present work is described in the next section, while the results are reported in section 3 and then discussed in section 4. The final section summarizes the conclusion, highlighting future development and limitations.

2. Materials and methods

2.1. Living lab description

An experimental tomato crop has been tested throughout the 2023 summer season in a living lab located in Parma, northern Italy (44°48′34″N 10°16′28″E). The living lab was developed by the University of Parma in collaboration with the Azienda Sperimentale Stuard within the framework of the National Research Center for Agricultural Technologies. Specifically, the Solanum Lycoperium L. Cv. HEINZ 1301 was grown according to the organic guidelines and specific instructions given by the involved research teams to carry out various experimental activities on different issues. In this perspective, a set of parameters were continuously monitored throughout the experimental season by a three-layer LoRaWAN network-based framework. Specifically, it provides irrigation recommendations based on a three-day weather forecast and data collected from the device layer installed in the living lab. Fig. 1 shows how the proposed system was characterized by an acquisition layer, which enables real-time data monitoring and data collection into a network server. An intermediate layer allows data to be processed and stored in cloud, where they can be used by user applications or processing nodes.

2.2. Acquisition data layer: LoRaWAN network and weather forecast API

The acquisition data layer was characterized by two different data sources. On the one hand, a few of end nodes (ENs) were installed in the field to collect data on soil and environmental conditions, and to manage

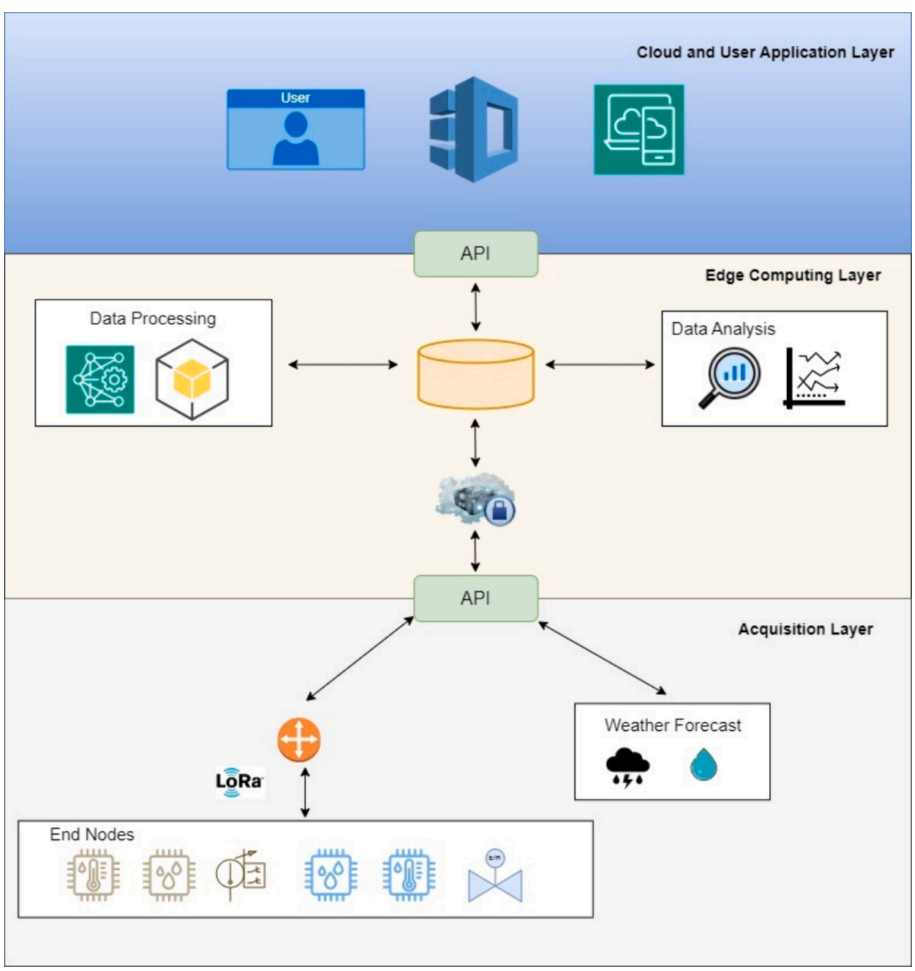

Fig. 1. Three interrelated layers architecture based on crop data, three-day weather forecast and machine learning algorithms.

the irrigation network. Specifically, signals were transmitted every 10 min over LoRa communication protocol between the end nodes and a gateway (GW) located in a warehouse in the vicinity of the experimental field. The latter communicated with a network server (NS) using an IP-based protocol that allows data storage and processing. A summary of IoT devices characterizing the end nodes block is provided below:

- Water meters to monitor water consumption (MClimate, 2023).
- Water valves for the collection of data on the status of the water supply and the automation of irrigation strategies (ON/OFF) (MClimate, 2023).
- 8-inch depth soil sensors to gain insights into the volumetric water content, temperature, and soil conductivity (MClimate, 2023).
- Environmental and weather sensors to collect weather and local environmental data (MClimate, 2023).

Concurrently, water management was optimized by aggregating real-time data with three-day weather forecasts. Specifically, an open-source web application programming interface (API) was used to collect hourly precipitation, relative humidity, and evapotranspiration reference rate (Open Meteo, 2024). The API provide a set of weather forecast based on various weather model dependent on the geographic region, such as the Integrated Forecasting System (IFS) for the European Union and COSMO 2I & 5 M for Italy (Bottazzi et al., 2021).

The API request has been formulated by setting the living lab coordinates, i.e., latitude = 44,1125 and longitude = 10,411, the European time zone (GMT + 1) and the reference period starting from July 28th to September 3rd, 2023.

2.3. Water management framework

2.3.1. Data analysis

Using Python Pandas library, a data set having 16,187 samples for each parameter was implemented by removing redundant and inaccurate data. The raw data was scrubbed for errors, duplicates or missing data, providing high quality information for further analyses and processing. The implemented dataset is described in Table 1, which reports the main statistical metrics.

Statistical methods were used to examine patterns among the data collected. Specifically, the associations between the measured variables were identified by calculating two different correlation coefficients. First, the Pearson correlation coefficient (r) was evaluated according to equation (1) to quantify the strength of the linear relationship between the parameters and to identify the corresponding direction. In this case, x_i is the value of an analyzed variable and yi represents the value of another one. At the same time, x_m and y_m are the corresponding mean values.

Furthermore, the correlation between the ranks associated to the raw data was explored by evaluating the Spearman correlation coefficient (r_s) according to equation (2), where d represents the differences between the ranks for each item and n is the size of the samples analyzed. Data analysis represents a crucial step to provide high quality information to make predictions.

$$r = \frac{\sum (x_i - x_m)(y_i - y_m)}{\sqrt{\sum (x_i - x_m)^2 (y_i - y_m)^2}} \tag{1}$$

$$r_s = 1 - \frac{(6 \sum d^2)}{n(n^2 - 1)} \tag{2}$$

2.3.2. Processing node: irrigation policy management

The amount of water available to the plant held in the soil ranges from a lower limit, known as the permanent wilting point, to an upper limit, known as the soil capacity. The magnitude of this range is highly dependent on soil structure and texture. Specially, the soil is characterized by macropores and micropores, which regulates the drainage flow and the amount of water available for the plant, respectively (Skovdal Christiansen et al., 2004). Irrigation or precipitation fills micropores that supply water to the plant. Meanwhile, water drains out of the macropores by gravity, trapping air. From this perspective, when the soil moisture exceeds the field's capacity, overwatering occurs because the water percolating to the lower layers of the soil cannot be used for the growth (Warrick, A.W., 2001).

The proposed framework provides an irrigation policy management based on the significant parameters resulting by the preprocessing steps. Specifically, over irrigations were performed to study soil saturation at the depth of tomato root. Water content in the soil has been monitored by the IoT soil sensors. As shown in Fig. 2, soil moisture declines at a faster rate after the irrigation event ends up to the field capacity point. From this point, the water content decreases more slowly because the macropores are already drained and the water from the micropores is withdrawn by evapotranspiration, which includes evaporation and transpiration. The first is the percentage of water that evaporates from the soil surface and the capillary fringe of the groundwater. The second is the process by which plants release water to the atmosphere due to their metabolic activities. The above-mentioned API provide the hourly reference evapotranspiration evaluated according to the Penman – Monteith method as set in (Pereira et al., 2015). It represents a recognized evapotranspiration rate calculated by considering clipped grass under given parameters. Hence, the specific crop evapotranspiration can be calculated by multiplying the reference value by the crop coefficient, which reflects the different contribution given by each specific crop. According to equation (3), reference evapotranspiration identifies the local climate and the crop drought conditions by considering the net surface radiation R_n, soil heat flux density G, the mean hourly air temperature at 1.5 to 2.5 m T, mean hourly wind speed at 2 m height u_2, pressure vapor saturation e_s at 1.5 to 2.5 m height, the mean actual vapor pressure at the same height e_a, the slope of the pressure vapor saturation vs temperature Δ, psychrometric constant γ, and two constants related to the crop type C_n and C_d.

Table 1
Statistical metrics of the dataset used to implement the predictive system.

	End Nodes Data							Weather Forecast		Crop Data
	Irrigation (ON/OFF)	Soil Moisture [RH%]	Soil Temperature [°C]	Soil Electrical Conductivity	Daily Hour	Environmental Temperature [°C]	Environmental Humidity [RH %]	Rainfall [mm]	Environmental humidity [RH %]	Evapotranspiration [mm]
Count	16,187	16,187	16,187	16,187	16,187	16,187	16,187	16,187	16,187	16,187
Mean	0.59	23.28	23.27	277. 84	11.49	25.66	56.07	0.098	56.25	0.177
Std	0.49	4.2	1.43	148.42	6.93	5.1	17.65	0.067	17.81	0.20
Min	0	16.45	19.6	58	0	14	18	0	18	0
25 %	0	20.11	22.4	148	5	22	42	0	42	0
50 %	1	22.66	23.4	261	11	25.1	55	0	55	0.5
75 %	1	25.51	24.2	375	18	29.5	69	0	70	0.33
Max	1	41.26	27.1	1484	23	38.9	96	10.6	96	0.67

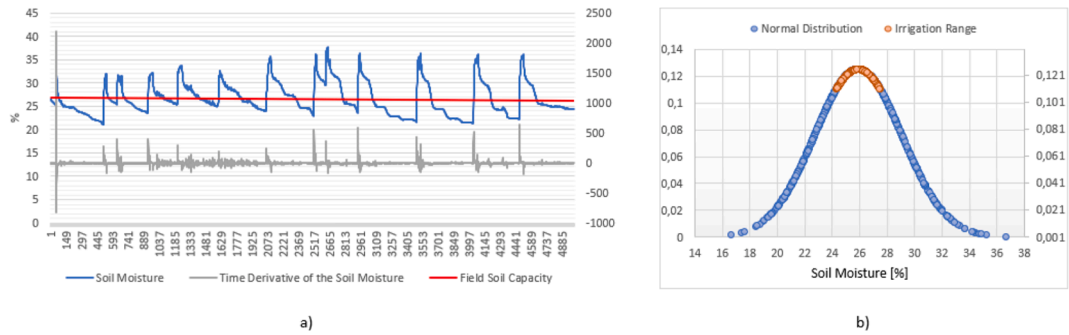

Fig. 2. a) Soil moisture properties derived from the data acquisition layer. The estimated soil capacity level is highlighted by the red line. b) Confidence interval used to drive irrigation decisions, where the lower and upper boundaries are calculated by subtracting and adding the half of the standard deviation to the corresponding mean value.

$$ET_{ref} = \frac{0.408\Delta(R_n - G) + \gamma(C_n/T + 273)u_2(e_s - e_a)}{\Delta + \gamma(1 + C_d u_2)} \quad (3)$$

As a result, the soil field capacity was considered as a reference value to guide the irrigation policy management, since it represents the limit to avoid overwatering. Specifically, for its experimental evaluation, the time derivative of the soil moisture was calculated to identify the zone where the rate of decline is slower after an irrigation event. As shown in Fig. 2, the resulting data were filtered for evaluating statistical metrics, such as mean (μ) and standard deviation (σ) to design a confidence interval for an irrigation framework based on soil capacity point. This interval focused on a limited region of the normal distribution by calculating the lower and upper confidence limits by subtracting and adding the half of σ to μ, respectively.

Irrigation and precipitation events influenced the amount of water retained by the soil. For the former, irrigation timing and volume affected water delivery. In the proposed living lab, the irrigation network was characterized by ON/OFF control valves that allowed to deliver a constant flow rate, close to 300 liters per hour (Preite et al., 2023a). In this perspective, the timing was the only parameter used to control the amount of water available to the plant.

The proposed approach also includes the rainfall contribution by exploring the three-day weather forecast. The main purpose is to define a reference value that would allow non-irrigation when the predicted event will have a significant saturating effect on the soil. Precipitation amount is expressed in millimeters, which corresponds to the volume of water (expressed in liters) per square meter of water on the surface. The reference value was calculated considering the amount of water

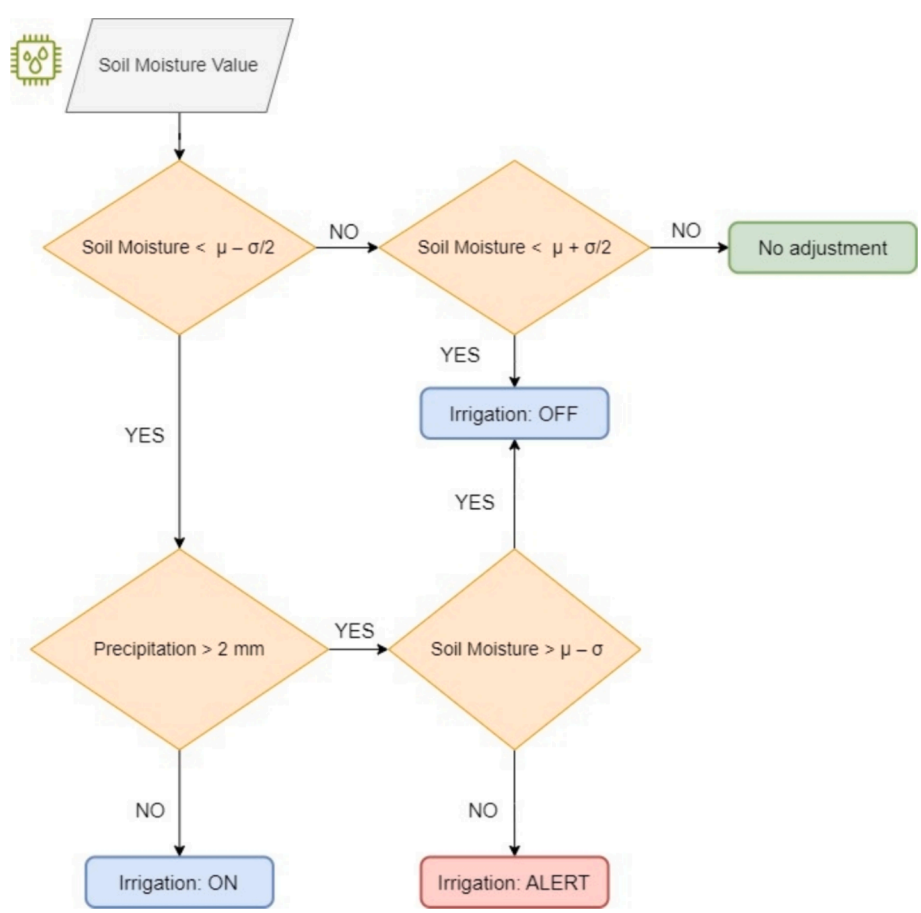

Fig. 3. Flow diagram of the adopted irrigation framework, which defines four different irrigation status: on, off, no adjustment and alert.

distributed by the irrigation system. Specifically, 300 liters of water per hour was delivered to an area of 132 square meters, almost 2 mm of water. As a result, if the predicted precipitation is higher than 2 mm the irrigation is not activate in the next days. In this case, the soil moisture is anyway monitored to avoid significant drought stress to the crop by considering a lower confidential limit of $\mu - \sigma$. An alert message is sent for manual handling when the soil moisture falls below this critical value. The adopted irrigation framework is summarized in Fig. 3.

2.3.3. Irrigation predictive algorithms

Predictive algorithms were developed with Python using scikit-learn library to enhance the sustainability of the irrigation management. Specifically, three machine learning algorithms were trained using the data set implemented according to the proposed irrigation framework. In this case, the issue deals with a classification problem, where four classes label are predicted starting from proper input data:

- Irrigation OFF = 0, to close the control valve and turn off the watering when the soil moisture exceeds than the upper confidential limit previously introduced (3467 records in the dataset).
- Irrigation ON = 1, to turn on irrigation on by moving the control valve when the soil moisture falls below the lower confidential limit (8875 records in the dataset).
- No adjustment = 2, if the watering status is not going to be changed when the soil moisture falls within the confidence range designed (3780 records in the dataset).
- Alert = 3, a warning message requiring farmer actions, when precipitations are detected but the soil moisture is below a critical value (65 records in the dataset). In this framework, the critical value has been defined as $\mu - \sigma$.

For the sake of simplicity, the output labels were changed to numerical values during the training and testing phases. Three models were trained by using three supervised classifiers, i.e., k-nearest neighbors, support vector machines, and artificial neural network. To train and test each model, the database resulting from the pre-processing was split into two datasets following the rules 70/30, respectively. Usually, a standardization of the input data is recommended to train the mentioned algorithms, since they can be truly sensitive to the scale of the input data. Since, the performance yielded by working with the original dataset was almost identical to that obtained with the z-score normalized dataset, this issue is not significant. As a result, the data were processed in the non-normalized format to give predictions directly in a meaningful measurement.

K-nearest neighbors stores the data available in the training dataset and classifies new data based on the feature similarity given by the majority votes of its k neighbors in an n-dimensional space (Kramer, 2013). For that purpose, the nearest neighbors of an unknown point are defined by computing the Euclidean distance from all the points in the dataset. In this case, k is one of the hyperparameters that needs to be tuned to maximize the accuracy of the model. The other is the weighting function used to make predictions, which can take on two options, uniform, and distance. The uniform function assigns the same weight to each neighborhood. Meanwhile, the associated weight is inversely proportional to the distance.

Support vector machine categorizes unknown data by considering the optimal hyperplane corresponding to the maximum distance, d, between the support vectors in each class. Indeed, support vectors represent the data points closest to the hyperplane, which are the only parameters influencing the final decision, unlike some other machine learning algorithms such as artificial neural network. A proper weight W is assigned to the identified support vectors and predicts the output by an optimization problem of the maximum distance d. As the real data are commonly noise, the classifier cannot tolerate outliers and fails in optimizing the maximum distance. In this case, a variable ξ is added to the constraints represented by the support vectors to consider the outliers. The right value of ξ is evaluated by using C as regularization parameter (Lasso regularization). According to equation (4) smaller C emphasizes the influence of ξ resulting in wider margin that can cause misclassification, meanwhile, larger C reduced the impact of ξ resulting in smaller margin with a less tolerance to outliers (Awad, M., et al., 2015).

$$C \sum \xi_n \tag{4}$$

If the data cannot be linearly separable, kernel functions are used to move through another dimensional space to define the better hyperplane. In the proposed framework, the model has been trained by considering the gaussian kernel function, which has one parameter γ. According to equation (5) small value of γ results in a linear behavior and large value can fall in overfitting. The model has been implemented by tuning the hyperparameters C and γ.

$$\mathrm{rbf}(\mathrm{Xi}, \mathrm{Xj}) = \exp(-\gamma \|X_i - X_j\|^2) \tag{5}$$

An artificial neural network has been trained by using back-propagation learning rules to classify new watering recommendations. Specifically, a supervised multi-layer perceptron neural network has been used to learn a non-linear function providing a classification output based on the input features (Zou, J., et al., 2008). Two different solvers were tested to train the classifier, i.e., stochastic gradient descent (sgd) and adaptive moment estimation (adam). Hidden layers perform the main computational part of the model, as they transform the values transferred from the previous or input layer by a weighted linear combination plus a bias and then computing a non-linear activation function. In this context, the lower bound of the hidden layer sizing was evaluated based on the framework provided by (Kuri-Morales, 2017) and it was used for test different combinations. Hence, a single layer with two, or six, or twelve neurons and a three layers layout with six, twelve, and six neurons were considered to perform the cross-validation process. Furthermore, two activation function associated to the hidden layers, i.e., hyperbolic tangent function (tanh) and the rectified linear unit (relu), were tested to define the better combination of the hyperparameters. The multi-label classification was finalized by applying Softmax activation function for the output layer, which is characterized by four neurons. Specifically, this function is a generalization of the sigmoid formula used for the logistic regression in binary classification and computes a normalized probability distribution of the values to determine the model output. The other hyperparameters were the initial learning rate and the regularization index. Lasso regularization was used to define the appropriate value of the weight associated with the values along the network. Fine tuning of the alpha parameters (regularization parameter) was crucial to avoid overfitting. At the same time, the initial learning rate controlled the weight updating during this process.

Each model has been improved through a heuristic approach. Indeed, several hyperparameter combinations have been automatically tested by using GridSearchCV, since there is no way to define in advance the optimal values that maximize the model performance. This function is provided by the scikit-learn library and evaluates the model performance via three-fold cross validation, which is a statistical method to compare the partitions involved in the analysis.

Once the optimal combinations had been defined, the models were validated against the test data by means of different metrics, such as confusion matrices and accuracy score. Specifically, three different parameters, i.e., precision, recall, and F-1 score, were evaluated by considering the resulting true positive (TP), true negative (TN), false positive (FP), and false negative (FN) indices. The first describes the number of samples that were correctly predicted. There were false positive and false negative when the predictions belonged to a different class in the real system. Finally, true negative was observed when a class was not correctly predicted. Table 2 reports the 4x4 confusion matrix resulted for each model, since the classification output was characterized by four different classes (i.e., OFF, ON, No adjustment, and Alert).

Table 2
4x4 confusion matrix used to assess the algorithms' performance.

ActualSamples	OFF	TN_{ALERT}	TN_{ALERT}	TN_{ALERT}	FP_{ALERT}
		TN_{NAD}	TN_{NAD}	TN_{ON}	TN_{NAD}
		TN_{ON}	FP_{ON}	FP_{NAD}	TN_{ON}
		TP_{OFF}	FN_{OFF}	FN_{OFF}	FN_{OFF}
	ON	TN_{ALERT}	TN_{ALERT}	TN_{ALERT}	FP_{ALERT}
		TN_{NAD}	TN_{NAD}	FP_{NAD}	TN_{NAD}
		FN_{ON}	TN_{OFF}	FN_{ON}	FN_{ON}
		FP_{OFF}	TP_{ON}	TN_{OFF}	TN_{OFF}
	No adjustment	TN_{ALERT}	TN_{ALERT}	TN_{ALERT}	FP_{ALERT}
		TN_{ON}	FN_{NAD}	TN_{ON}	TN_{ON}
		FN_{NAD}	FP_{ON}	TP_{NAD}	FN_{NAD}
		FP_{OFF}	TN_{OFF}	TN_{Off}	TN_{Off}
	Alert	FN_{ALERT}	FN_{ALERT}	FN_{ALERT}	TN_{NAD}
		TN_{NAD}	TN_{NAD}	TN_{ON}	TN_{ON}
		TN_{ON}	FP_{ON}	FP_{NAD}	TP_{ALERT}
		FP_{OFF}	TN_{OFF}	TN_{OFF}	TN_{OFF}
		OFF	ON	No adjustment	Alert
		Predicted Samples			

For better classification performance, TP and TN should be maximized and FP and FN should be the smallest. Thereby, precision ranges from 0 to 1 and describes what percentage of the specified class is actually true.

$$Precision_{CLASS} = \frac{\sum TP_i}{\sum TP_i + \sum FP_i} \tag{6}$$

Recall shows the true positive rate, which is the percentage of the predictions belonged to the given class.

$$Recall_{CLASS} = \frac{\sum TP_i}{\sum TP_i + \sum FN_i} \tag{7}$$

F1 – score considers both false positives and false negatives by computing the harmonic mean of precision and recall.

$$F1 - score_{CLASS} = \frac{2*Precision*Recall}{Precision + Recall} \tag{8}$$

Macro average and weighted average were also determined to take into account the influence of all classes. Thus, the former represents the arithmetic average of the precision of all classes, while the latter is based on the precision classes weighted by considering the number of samples contained in each class.

$$Macro - Average\ Precision = \frac{Precision_{OFF} + Precision_{ON} + Precision_{NAD} + Precision_{ALERT}}{4} \tag{9}$$

$$Weighted - Average\ Precision = \frac{Precision_{OFF}*N_{OFF} + Precision_{ON}*N_{ON} + Precision_{NAD}*N_{NAD} + Precision_{ALERT}*N_{ALERT}}{N_{TOT}} \tag{10}$$

Finally, the prediction capabilities are evaluated by computing the accuracy score according to equation (11), where y represents the real value and ŷ the corresponding predicted value.

$$Accuracy_{CLASS}(y, \hat{y}) = \frac{1}{n_{values}} \sum_{i=0}^{n_{values}-1} 1(\hat{y} = y) \tag{11}$$

The resulting decision regions of each model were evaluated by using the mlxtend library to visualize the performance achieved. Visualizing these regions truly shows the complexity of the model, highlights overfitting, and gives an indication of the prediction confidence level.

2.4. Water and energy savings

The proposed framework allows to regulate the soil moisture in a neighborhood of the capacity level to avoid percolation phenomena. In this perspective the potential water and energy savings have been probabilistically determined by performing a Monte Carlo simulation of 1000 tomato seasons. For any uncertain variable, this method leverages a probability distribution to develop a model of possible outcomes (Zio, E., 2013). Specifically, starting from a normal distribution characterized by the soil capacity level as the mean value and μ-σ/2 as the standard deviation, the resulting activation inputs provided by the prediction system were used to preliminarily evaluate a range of the potential water savings along a conventional irrigation period of 70 days (i.e., 100,800 min, 10,080 samples considering the operating frequency of the LoRaWAN network). An activation input results in an irrigation time of 10 min delivering nearly 300 liters/hour. In this way, the total water consumption has been evaluated and compared with the watering recommendation provided by the IRRIFRAME platform. Following the same framework, energy savings were evaluated by considering the power supply of the pump driving the irrigation system, which is 4 kW.

The output of the Monte Carlo simulation is a normal distribution of the water and energy savings. Specifically, an appropriate confidence interval with 95 % coverage probability has been provided by plus and minus two estimated standard deviation from the estimated mean value.

3. Results

3.1. Data distribution and correlation

Fig. 4 (a-b) reports the correlation matrices describing the strength and the direction of the correlations of the data collected according to Pearson and Spearman ranks.

As it can be seen before, the performed correlation analyses provided consistent results that allow to reshape the data frame to train and test the predictive algorithms. In this perspective, the parameters characterized by redundant correlations in strength and direction with the irrigation decision were dropped to avoid overfitting.

According to the proposed framework, the parameters were grouped into three classes considering the corresponding relationship:

- Soil properties.
- Environmental properties.
- Weather forecast parameters.

Soil properties and evapotranspiration had the most significant correlation. Considering the first one, moisture content showed the highest correlation with the irrigation decision. Electrical conductivity had a lower correlation than soil moisture in the same direction, so it was not selected for the final dataset. Although temperature was less correlated than soil moisture, it was also included because the correlation was in the opposite direction. Among the environmental characteristics, temperature and relative humidity were included because they had weak correlation and were in the opposite direction. Since humidity was consistent in direction with a weaker relationship, precipitation and evapotranspiration were the variables processed for weather prediction.

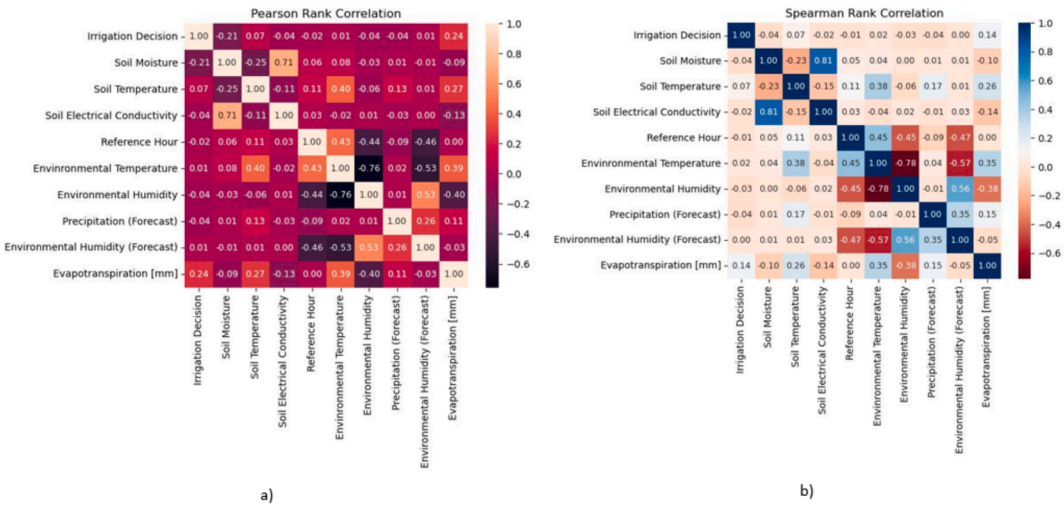

Fig. 4. a) Pearson rank correlation of the collected data. b) Spearman rank correlation of the collected data.

3.2. Processing node: Irrigation predictive algorithms

3.2.1. K-nearest neighbors' classifier

The cross-validation process allows the hyperparameters to be maximized. Specifically, the model was trained by considering 13-nearest neighbors and the weights are handled by the distance function.

3.2.2. Support vector machine classifier

Kernel radial basis function was implemented for the training phase. The regulation index C and the kernel function parameter γ were set to 1000 and 0.1, respectively.

3.2.3. Artificial neural network

Artificial neural network was employed by considering the three hidden layer configuration resulting from the cross-validation step. Hyperbolic tangent was the corresponding activation function. In this framework, Adam was the algorithms used with a constant learning rate and a regulation index of 0.01.

3.2.4. Algorithm's comparison

Confusion matrices and metrics are used to assess the performance in predicting field data. Overall, the three models were characterized by high accuracy, with the highest value of 99,61 % achieved by the artificial neural network. This was followed by the k-nearest neighbors and support vector machine with 99,46 % and 99,20 %, respectively.

True positives, false positives, false negatives, and true negatives are reported in the confusion matrices in Fig. 5 (a-b-c). In addition, the Fig. 5 (d-e-f) highlights the 2-D decision regions according by soil moisture content and rainfall forecast amount, taking into account a mean value for the other features involved.

The comparison between the classifiers based on the calculated metrics is reported in Table 3.

3.2.5. Water and energy savings

The water savings confidence interval resulting from the Monte Carlo simulation ranges from 14.5 % to 27.6 %. The lower and upper bounds for energy savings are 49.2 % and 57 %, respectively. As mentioned

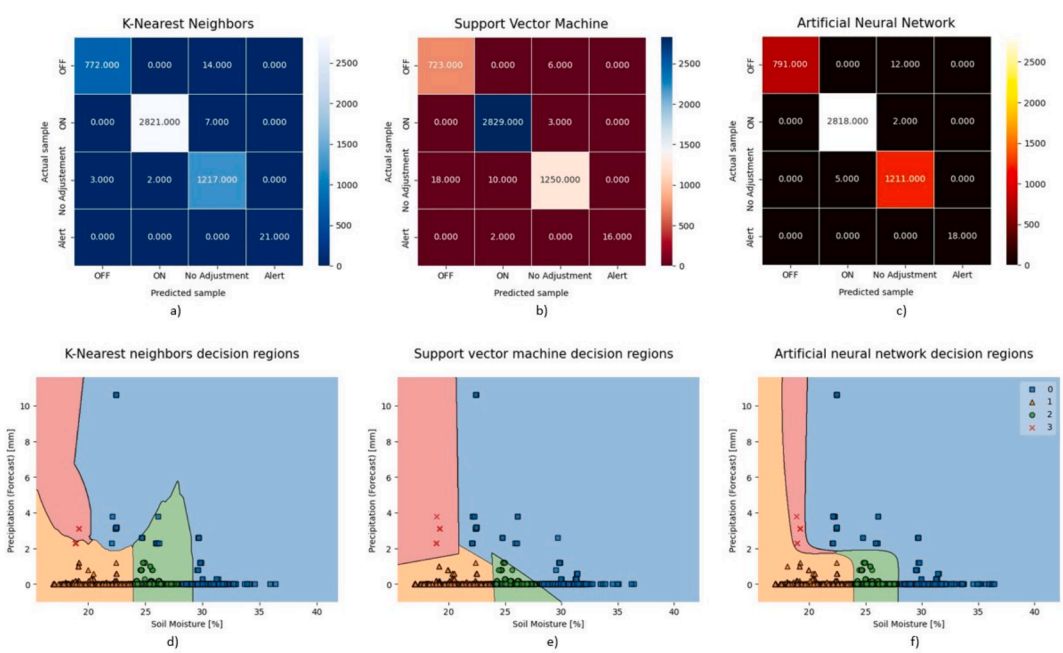

Fig. 5. a-b-c) Confusion matrices for the implemented machine learning algorithms. d-e-f) decision region boundaries for the implemented machine learning algorithms.

Table 3
Classification metrics for the implemented machine learning algorithms.

	Precision			Recall			F1-score			Samples
	KNN	SVM	ANN	KNN	SVM	ANN	KNN	SVM	ANN	
OFF	0,9961	0,9757	1	0,9822	0,9918	0,985	0,9891	0,9837	0,992	
ON	0,9993	0,9958	0,9982	0,9975	0,9989	0,9993	0,9984	0,9974	0,9988	
No Adjustment	0,9830	0,9929	0,9886	0,9959	0,9781	0,9959	0,9894	0,9854	0,9922	
Alert	1	1	1	1	0,8889	1	1	0,9412	1	
macro avg	0,9946	0,9911	0,9967	0,9939	0,9644	0,9951	0,9942	0,9769	0,9959	4857
weighted avg	0,9947	0,9920	0,9961	0,9946	0,9920	0,9961	0,9947	0,9920	0,9961	4857

earlier, these intervals cover 95 % of the estimated values.

4. Discussion

The data collected from the developed living lab were used to provide a three-layer predictive algorithms-based irrigation system. Specifically, information on soil and environmental properties were gained on 2023 tomato crop season by implementing an IoT LoRaWAN-based network, which transmits data every 10 min to the network server. These data were stored in database and processed in combination with the weather forecasts provided by an open-source API based on international recognized weather models.

The results show how the developed models achieve significant performance in predicting irrigation status. All the tested algorithms achieved comparable accuracy, close to 99 %. A classification problem characterizes the irrigation framework, which avoids overwatering by defining the soil capacity as thresholds to manage the water delivery. The outputs are four classes, i.e., OFF, ON, No Adjustment, Alert.

In order to train and validate the models, the data were arranged according to the correlation analyses results. Thereby, the Pearson and Spearman correlations yielded consistent results to determine the right combination of parameters to avoid overfitting and maximize the accuracy. A final dataset of 16,187 samples and 6 features was defined by considering soil water content and temperature, ambient temperature, and relative humidity as real-time attributes; precipitation amount and evapotranspiration reference rate for the next three days were included as forecast features. The resulting dataset was split 70/30 into training and test data. The training was performed by determining the optimal value of the hyperparameters with cross validation. Overall, the nearest neighbor of 13, the weight distance function, the high regularization term C, the resulting neural network structure, and the low value of the regularization parameter alpha reveal how the noisiness of the data was not significant.

Fig. 5 provide a powerful overview on the classification results. Decision region visualizations and confusion matrices highlight how there are significant differences between the models at nearly the same accuracy. Indeed, the multi-layer perceptron neural network define reliable decision boundary to associate the right class to unknown data, meanwhile the k-nearest function and support vector machine fail in differentiating some cases. This issue is also described by the evaluated metrics reporting in Table 3. In this perspective, a higher amount of false positive and false negative was reached for the k-nearest neighbors and support vector machine. Moreover, the warning alarm was correctly predicted by both neural network and k-nearest neighbors, while two false positives resulted from the support vector machine model.

Digging deeper into the results, the algorithm is used as a processing node to make predictions on the field data every 10 min. The main aim of the performed classification problem is to reduce the water consumption by avoiding overwatering during irrigations. Thereby, the scientific literature highlights how a percolation phenomenon occurs when the water content exceeds the soil capacity point. A potential water and energy savings are estimated by comparing these preliminary results with the irrigation volume recommended by IRRIFRAME. Therefore, the results of the Monte Carlo simulation outline how the resulting confidence interval ranges from 14.5 % to 27.6 % for water savings while side it ranges from 49.2 % to 57 % for energy savings.

This framework provides a cost-effective solution, as the authors of (Stefanini et al., 2023) assess the economic viability of the data acquisition layer implemented and state that almost 2 years are required to recoup the initial investment. The data collected by this layer are then aggregated with the weather forecast data collected by an open-source platform. The main key of the developed framework is the soil capacity evaluation. Once it is calculated, the dataset was adapted to it and the algorithm can make new predictions based on the new operating conditions. In this perspective, there are also some limitations. A wide range of soil types and thus different soil capacity level must be tested to assess the robustness of the model and the crop status data should also be aggregated to affine the model and account for the real water needs of the crop at different growth stages.

5. Conclusion

The agricultural sector accounts for almost 70 % of global water withdrawals, recording one of the major impacts on water scarcity issue. Several efforts reported in the scientific literature have proven how the 4.0 scenario is a powerful solution to improve agricultural sustainability and minimize water withdrawals. Therefore, there are several cutting-edge solutions, including blockchain, internet of things, digital twins, deep learning and machine learning algorithms, and other computer applications, for reducing wastewater and improving irrigation schedules.

The proposed work aims at analyzing the application of predictive algorithms in agriculture for irrigation purposes. Specifically, the scientific literature highlights how machine learning techniques are widely applied in various disciplines because they offer highly scalable solutions capable of automating processes based on predictions. In addition, the scientific literature describes how the proposed applications in agriculture focus on three different levels, i.e., pre-harvest, harvest, and post-harvest. In this perspective, the first level aims to maximize the effectiveness of the growing conditions, which also include irrigation management.

In the proposed framework an artificial intelligence application based on a three-layer architecture (i.e., data acquisition, edge computing, and cloud layer) has been investigated to address water scarcity. The main purpose is to improve the irrigation schedules to avoid overwatering, which causes percolation of the water from the soil to the lower layers when the water content exceeds than the soil capacity. In this perspective, the water moves back to the groundwater table, but it cannot be used directly for the plant growth. The soil capacity point has been experimentally assessed by defining a Gaussian distribution of the soil moisture having given mean and standard deviation values, which are evaluated by calculating the time derivative of the soil moisture to identify the area where the rate of decline is slower after an irrigation or a precipitation event. As a result, an irrigation confidence interval has been implemented to create four different classes (i.e., irrigation OFF, irrigation ON, no adjustment, and alert) that are used to set up a multi-classification problem.

To address this issue, a dedicated living lab monitored the growing conditions of a tomato crop throughout the whole season. The core part of the data acquisition layer is an IoT network, characterized by several

end nodes installed in the field, which collect information on environmental and soil conditions every 10 min. The collected data are transmitted via LoRa communication protocol to a gateway, which exploits an internet protocol to communicate with the network server. At this point, the data can be stored into databases and processed by scrubbing for errors, duplicates, or missing data, and finally aggregated with the three-day weather forecast data provided through an open-source API. The resulting dataset, characterized by 16,187 samples and 12 features (i.e., irrigation decision, soil moisture, soil temperature, soil electrical conductivity, daily hour, ambient temperature, ambient humidity, rainfall, environmental humidity forecasted, and evapotranspiration), has been used to gain insights into features correlation by means of Pearson and Spearman correlation coefficients. Following this path, redundant parameters are removed to create a consistent dataset for training and testing the machine learning algorithms.

Deepening into the results, three different classifiers, i.e., multi-layer perceptron neural network, support vector machine, and k-neighbors classifier have been trained and tested by splitting the resulting dataset into 70 % and 30 %, respectively. Prediction performances have been assessed by evaluating boundary decision regions, confusion matrices, and various metrics, such as accuracy, precision, recall, F1-score, macro-average precision, and weighted-average precision. Despite on the three classifiers work with high comparable accuracy (close to 99 %), multi-layer perceptron neural network is the more suitable solution because it minimizes false predictions as it is shown by the confusion matrix and revealed by the boundary decision regions. The highest levels of macro and weighted average precision also support this finding.

As the final step, the performance of the proposed framework has also been investigated by evaluating the resulting water and energy saving. Therefore, a Monte Carlo simulation shows that water savings of up to 27.6 % and energy savings of up to 57 % can be achieved compared to a conventional tomato season.

Summing up the results, the predictive algorithm-based irrigation system is a cost-effective solution to reduce water withdrawals by avoiding soil percolation phenomenon. It can be scaled to other crops, as the core part of the system is the reliable determination of the soil capacity point, which affects the confidence interval implemented in the preprocessing stage. Future research should further investigate the robustness of the model by evaluating a wide range of soil types and aggregating crop data, such as chlorophyll content or sap ion concentration to also assess plant drought stress.

CRediT authorship contribution statement

Luca Preite: Conceptualization, Methodology, Validation, Writing – review & editing, Investigation, Formal analysis, Visualization, Resources, Software, Writing – original draft. **Giuseppe Vignali:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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